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A cradle-to-customer life cycle assessment case study of UK vertical farming

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ABSTRACT

Global population rise, increased rural to urban migration, climate change and conflict have placed strain on global trade and food supply chains. Vertical farming (VF) is a relatively novel method for cultivating many fresh fruits and vegetables, and often is claimed to have lower environmental impacts than field cultivation. However, to date there are few studies utilising primary data which have evaluated the environmental impact VF may have. This study utilised Life Cycle Assessment (LCA) to evaluate the environmental impact of lettuce production in a commercial vertical farm in the UK. This was compared with data from the literature on the impacts of lettuce in field cultivation. Scenarios were also examined to evaluate the impact of VF systems with varying electricity sources, and waste, water and nutrient management options. The VF was found to have a similar or lower Climate Change impact than field cultivation, depending on energy source used. VF had a slightly higher environmental impact than field cultivation in some other environmental impact categories such as freshwater eutrophication potential and acidification potential. Electricity demand and the medium used for 'plugs' in the VF were found to be the main hotspots of the system. The results of this study provide insights on the environmental impact and resource efficiency of VF, feasibility for larger scale deployment, and provide further empirical evidence to support the claims made previously in the literature.

1. Introduction

1.1. The need to transform current food systems

As the global population continues to rise, questions on food and energy security are becoming increasingly critical. According to the United Nations (UN), the world's population is predicted to increase by two billion (bn) people by 2050, from 7.7 bn to 9.7 bn, and peak at nearly 11 bn by 2100 (United Nations, 2019a). This predicted growth rate has highlighted a clear need to develop, adapt, and change current global production systems to accommodate more people with increased demand for resources and services, including needing 56% more food by 2050 (from 2010 levels) (Rangathan et al., 2018). Migration to urban centres is also predicted to increase, with a further 2.5 bn more people living in cities or urban areas by 2050 (approx. 2/3 of the global population) (United Nations, 2019b).

Rural to urban migration is accompanied by other changes which have also impacted global food and energy security. In recent years, threats to agriculture and food security have been exacerbated by drought and land degradation due to climate change (Webb et al., 2017), and conflicts such as the Russia-Ukraine war, and the Covid-19 pandemic have strained global trade and food supply chains (EC, 2023). Total productive agricultural land area is also dwindling. These pressures have prompted innovation to create technologies which can grow more food on smaller land areas, a notable one being the introduction of vertical farming (VF).

Anthropogenic greenhouse gas (GHG) emissions from the food sector account for 21–37% of total global GHG emissions; from agriculture and land use, storage, transport, packaging, processing, retail and consumption (IPCC, 2022). Estimates from crop and livestock activities are 9–14%, land-use and land-use change 5–14%, and the transport and supply chain activities accounting for 5–10% (IPCC, 2022). The emissions from the food sector are significant, and avenues to reduce the environmental impact of these systems are needed. Between 2010 and 2016, global food loss and waste equalled 25–30% of total food produced, which is the equivalent of 8–10% of total anthropogenic GHG emissions (IPCC, 2022). Reducing this loss is imperative, and VF may be an appropriate avenue to explore, given its ability to shorten the supply

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chain between producer and consumer, and to ensure consistent production of crops for an increasingly urban, and growing population.

Improvements to total agricultural productivity and farming techniques have increased since the 1970s (an average of 2.5% growth per year), with Total Factor Productivity (TFP-the measure of the amount of agricultural outputs produced from the combined set of land, labour, capital and material resources employed in farm production) now replacing resource intensification as the primary source of growth in world agriculture (Morgan et al., 2022; USDA, 2023). Though agricultural productivity has increased globally, it has been at varying rates, with developing economies and large agricultural-producing lands such as Mexico, China and Brazil having some of the highest average annual growth in TFP from 1991 to 2020 (Morgan et al., 2022). Within the UK, great improvements in TFP have occurred, 58% since 1973, yet the UK still imports 46% of food consumed by economic value (DEFRA, 2021). The UK has become increasingly reliant on imports for fruit and vegetable demands, importing 47% of vegetables and 84% of fruit (English and Hughes, 2022). VF may be an avenue to tackle some of these supply issues to the UK.

1.2. A brief background to vertical farming

VF is a form of Controlled Environment Agriculture (CEA), which is a grouping of technology-based crop production systems, comprised of various nutrient delivery systems (hydroponics, aeroponics, aquaponics etc.) in order to optimise the use of resources such as water, energy and space (Gargaro et al., 2023; Kozai and Niu, 2015). In CEA, crops are farmed with optimised growth conditions, avoiding the impact of external factors. New innovative farming methods such as VF are now providing even more benefits by using the vertical plane to maximise the use of land area as well. By utilising 'stacked' production systems, VF can minimise land use, and help mitigate the diminishing available arable land.

Typical VF set-ups boost yield by using artificial lighting (now commonly LEDs) to deliver precise lighting spectra for particular crops (Orsini et al., 2020), and soilless nutrient delivery systems (e.g. hydroponics) to grow plants in circulating water enriched with metered quantities of nutrients (Birkby, 2016). A major benefit of VF is reduced water consumption thanks to the use of closed-loop recycling systems, with some estimates suggesting water use can be up to 90% less than conventional farming (Barbosa et al., 2015). VF also increases yields per land area, as evidenced by Touliatos et al. (2016), where, even compared to a horizontal CEA system, VF increased the number of plants per m^2 (from 50 to 1000) and fresh weight yield per m2 from 6.9 kg FW m^{-2} to 95 kg FW m^{-2} (Touliatos et al., 2016). However, though VF does have significant potential, the high energy demands tend to make cultivation expensive (and potentially environmentally costly), with average prices for lettuce from vertical farms over twice that from field production (Nicholson et al., 2020).

Previous LCA papers on VF in the UK have been published, notably Sandison et al. (2023) and Casey et al. (2022). Both papers have evaluated the lifecycle implications of VF in the UK context, yet neither utilised primary cultivation data from commercial farms in the UK. Martin et al. (2023) and Blom et al. (2022) have conducted LCA's on VF using primary commercial farm data, but in the Swedish and Dutch context respectively, which can offer insight into the impacts of VF in the UK context, but inevitably misses much of the nuance which comes with modelling impacts with country specific data. The results and insights from these papers will be included in the discussion section.

1.3. Aims

To date, there have been very few reported Life Cycle Assessments (LCA) of VF that employ exclusively primary data from the practice of commercial farms in their calculations. The aim of the present research was to address this gap by undertaking a LCA of lettuce grown in a commercially operated VF in the UK from the delivery of inputs to the farm, through cultivation and harvesting, to delivery of a lettuce crop to its point of use in the restaurant market. Using the Life Cycle Inventory (LCI) data of the case-study farm, an attributional Life-Cycle Assessment (aLCA) was carried out following the principles of ISO14040 and 14044 (Finkbeiner et al., 2006; ISO, 2006a,b).

This scope encapsulates all operational facets of a VF set-up (initial construction excluded) and is used to identify the hotspots from production where the environmental impacts are highest. Lettuce was selected as the crop to evaluate, as it is one of the most commonly cultivated within these systems. The work is focussed on understanding the environmental impacts of commercial-scale VF, and on understanding how the environmental impacts from commercial VF compare with those for conventional field-cultivated lettuce. Scenarios for the use of current and future electricity projections, as well as water and nutrient recycling, alternative waste processing and alternative packaging are explored to evaluate further process development potential for VF.

2. Methodology

2.1. LCA methodology

Environmental LCA is a systems-analysis tool to display the potential environmental impact profile of products and services across a range of environmental impact categories. The methodological framework used in this paper is that of the ISO standards 14040 and 14044 (ISO, 2006a, b).

The following VF scenarios were examined:

- aLCA of lettuce produced by VF at commercial scale in the UK.
- Alternative energy scenarios (current UK grid (2022), UK 2030, 2040, 2050 grid mix projections).
- Alternative nutrient recycling scenarios (Re-circulating system).
- Alternative waste-management scenarios (Anaerobic digestion and industrial composting).
- Alternative packaging scenario (single use cardboard crates and plastic bags).

2.2. Goal and scope

The goal of this work is to carry out LCA research on the environmental impacts of a commercial vertical farm growing lettuce in the UK. The outputs of the work are intended to inform an understanding of the feasibility for deployment of VF for food cultivation in the UK context, and how its impacts compare with current business-as-usual field cultivation. The study is focussed on the cultivation of a lettuce crop within the vertical farm, and all other inputs, outputs and transportation needed to fulfil this.

2.2.1. Functional unit

The functional unit (FU) for all scenarios is:

- The use of a commercial VF system to produce 1 kg of marketable lettuce delivered to the customer.

This FU represents the lettuce being cultivated and the transport needed for delivery to restaurant customers. For the purpose of comparison, this FU is also used for a conventional, field-grown lettuce crop and its delivery.

2.2.2. System boundaries

This cradle-to-customer assessment encompasses the whole lifecycle of the vertical farm evaluated, including the environmental impact of producing inputs for the system (e.g., the emissions of producing, transporting and utilising jute fibre plugs). It encompasses the delivery of products to the system, their utilisation in cultivation, the waste leaving the system, and the lettuce products being delivered to the customer (in this case, partner restaurants). Assumptions regarding the environmental burdens from the use and disposal of lettuce from the customer are assumed to be equivalent to those of existing food operations and hence were not modelled. The system boundary therefore starts with the production of inputs and ends with the delivery of produce to customers. Fig. 1 shows the system boundaries of the vertical farms lettuce production.

Included within the system are:

- Delivery of inputs to the system, and the impact associated with their production.
- Production of lettuce within the VF, including packaging and cleaning operations.
- Management of wastes. (mass and composition of the outgoing waste is based on data provided by the farm and waste collection company used and includes municipal solid waste and recycling. Supplementary Material give the assumptions used in waste modelling)
- Transport of lettuce crops to customers.

Excluded from the system are:

- The use and disposal of lettuce at the customers and by their consumers.
- Embodied emissions from capital goods (equipment, VF buildings).
- Human labour and employee transport.

2.3. Data, data quality and impact assessment

Ecoinvent v3.8 (Ecoinvent, 2022) was the main database used for the LCI and emission factor data. The primary inventory data for the hydroponic VF system was obtained by consultation with the UK-based VF company. Foreground data was combined with background data in the LCA software SimaPro v9.3.0.3 (Pre Consultants, 2021) to compile and calculate the LCA.

The Life Cycle Impact Assessment (LCIA) was conducted using the Product Environmental Footprint approach of the European Commission (EF 3.0 in SimaPro v9.3.0.3). The following six impact categories are presented in detail, with the remaining 22 impact categories in EF 3.0 provided in Supplementary Material.

- Climate change
- Acidification
- Freshwater Eutrophication
- Land-Use
- Fossil resource use
- Water use (EC-JRC, 2018; Fazio et al., 2018).

Based on experience and previous research, these six impact categories were considered to be the most appropriate to focus upon regarding the environmental impacts from agricultural systems (controlled environment or open-field) (Bartzas et al., 2015; Dias et al., 2017; Martin et al., 2023). Regarding land use, the ReCiPe 2016 v1.1 midpoint (Heirarchist perspective) approach (ReCiPe 2016 Midpoint (H) in SimaPro v9.3.0.3) was used to complement the EF 3.0 approach. EF 3.0 presents the results for Land-Use in a dimensionless unit which combines area with soil impacts, whereas ReCiPe presents land use as a simple area per annum (m^2a) and this indicator was used for comparison of the land occupied for the Functional Unit between VF and field cultivation.

2.4. Life Cycle Inventory (LCI) analysis

Fig. 1 shows the detailed flow of inputs and outputs to the current practice system at the vertical farm.

2.4.1. Vertical farm

The input and output quantities for the system are given in Table 1 and subsequent sections. The area of the growing facility is 150 m^2, all contained in one controlled environment room, housing hydroponic growing systems with 4 layers of stacking. The overall land area for all processes on the vertical farm site is 238 m^2, which include offices, packrooms, storerooms etc.

The VF process evaluated has six main stages: delivery of inputs to the system, cultivation of lettuce crops, harvesting and packaging, cleaning and disposables, outputs from the system, and delivery of the outputs. The cultivation stage accounts for 91% of the electricity demand and 75% of the water demand.

Seeds were sourced from a specialist UK supplier and their production and sourcing emissions are included in the analysis, as are the emissions associated with nutrients and fertilisers used in the system (also sourced from a UK based company). Harvesting and packaging inputs include all the packaging materials needed to package and deliver

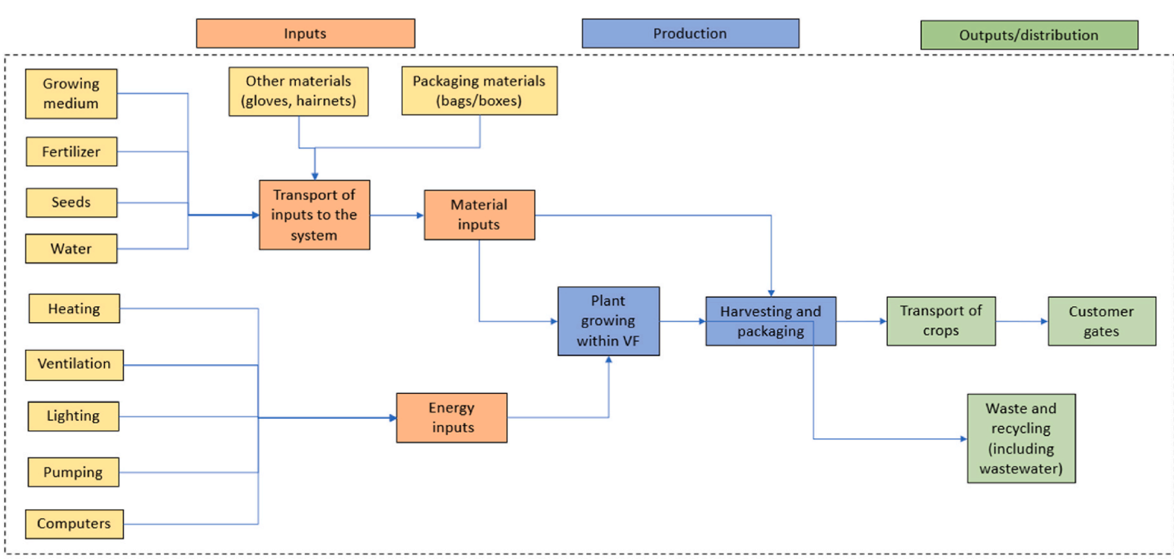

Fig. 1. System boundary for the cultivation of lettuce within a vertical farm in the UK, from cradle-to-customer.

Table 1
LCI data for the VF set-up evaluated. (Inputs and outputs normalised to the Functional Unit).

Stage	Flow	Process/activity	Value	Units
Transport of inputs to the farm	Input	Seed delivery	0.00092	kg km^{-1}
		Chemical and nutrient delivery	0.0088	kg km^{-1}
		Grow media and plug delivery	0.0094	kg km^{-1}
		Other inputs	0.005	kg km^{-1}
Cultivation stage	Input	Seeds	0.0005	kg
		Calcium Nitrate	0.008	kg
		Magnesium Sulphate	0.008	kg
		Solufeed F	0.020	kg
		TEC	0.00066	kg
		Nitric Acid (25%)	0.034	kg
		Hydrogen Peroxide (50%)	0.045	kg
		Grid electricity (all stages)	17.25	kWh
		Water (all stages)	0.028	m^3
		Jute fibre plugs	0.063	kg
		Bin liners	0.043	kg
Harvesting and packaging	Input	Reuseable plastic crate	0.003	kg
		Labels	0.00068	kg
Cleaning and disposables	Input	Surface disinfectant	0.0014	kg
		Surface detergent	0.0014	kg
		Bactericidal washing liquid	0.0021	kg
		Latex gloves	0.0010	kg
Outputs	Output	Lettuce to customer	1.00	kg
		Waste Lettuce	0.075	
		General waste	1.35	kg
		Recycling waste	0.035	kg
		Wastewater	0.035	m^3
		Nutrient waste (all nutrients by element)	0.030	mg l^{-1}
		Nutrient waste (all nutrients by element)	0.031	kg
Transport of outputs from the system	Output	Delivery to customer	0.0067	kg km^{-1}
		Waste collection	0.0096	kg km^{-1}

the lettuce. For deliveries to the customers, reuseable plastic crates are used. Alternative packaging scenarios were explored in this research to reflect typical packaging for produce going to retail (biodegradable polybags and cardboard crates for small and large deliveries respectively). Cleaning and disposables include the cleaning products and protective equipment needed to maintain cleanliness levels. Full detail on the LCI datasets and assumptions used are given in the Supplementary Materials.

2.4.2. Electricity

Table 2 gives the electricity demand for the system, calculated from 'hours on' and 'wattages' for each component. The main electricity demand was from the lighting (LEDs) and the heating, ventilation, and air conditioning (HVAC) systems at 56% and 33% respectively. Office space contributed approx. 10% to the overall electricity demand.

The electricity for the VF company is provided currently via a 100% renewable energy guarantee from the supplier, this is modelled as wind energy. A scenario of electricity supply from the current UK electricity average grid mix (2022) was also modelled to compare the environmental impact of the VF's specific electricity supply to current UK average electricity supply. For the analysis of VF potential in the UK, national electricity grid mix projections for 2030, 2040 and 2050 are also modelled.

Table 2
Breakdown of electricity use, by component. (All values normalised to the Functional Unit).

Component	Value	Units	Percentage of total (%)	Hours on per day (x/24 h)
Total electricity	17.25	kWh	100	–
Lighting	9.68	kWh	56.10	18
Pump	0.1	kWh	0.58	4.4
UV lamp	0.01	kWh	0.058	4
Water heater	0.12	kWh	0.69	2
Fridge	0.0095	kWh	0.055	24
HVAC	5.58	kWh	32.55	24
Dehumidifier	0.1	kWh	0.60	8
Pressure washer	0.012	kWh	0.067	0.21
Other (Office)	1.64	kWh	9.26	–

2.4.3. Wastewater

Nutrients input to the system are categorised by chemical (calcium nitrate etc, see Table 1). Elements or ions are used for wastewater outputs. The measured composition of wastewater sent to the sewer is given in Table 3. All of the elements in the wastewater are below maximum threshold limits and are not of concern to the local UK water management company.

2.4.4. Solid waste

Waste collection is carried out by a waste contractor, based within the local area of the vertical farm. General waste is sent to incineration, and recyclate is sent to a waste sorting and recycling centre. General waste and recycling composition is given in the Supplementary Material. General waste almost exclusively includes roots and plug material (biogenic), with very small contributions from other sources (paper towels, gloves). The paper based recyclate is composed mostly of cardboard waste from deliveries of consumables, as well as some office wastepaper. The plastic recyclate is from deliveries, as well as plastic film wrap, hairnets etc.

The weights provided by the waste collection company were found to be overestimated, so trials were performed and adjustments calculated (see Supplementary Material).

2.4.5. Transport

The LCI includes the transport of inputs and outputs of the system. The inputs are typically bulk ordered to the farm and stored onsite. Values for the transport of inputs were calculated by assuming the total amount of input needed per year, multiplied by the distance travelled from the supplier to give a value in kg km^{-1}, normalised to the Functional Unit.

The outputs from the system include the delivery of lettuce to the customer, and waste collection. Waste collection occurs weekly, and is of a consistent mass, hence the kg km^{-1} from waste collection can be

Table 3
Wastewater composition per litre.

Element/Chemical	Value	Units
Nitrate in water (NO_3)	1004.25	mg l^{-1}
Total Sulphur (SO_4)	525.49	mg l^{-1}
Soluble Molybdenum	0.15	mg l^{-1}
Phosphorous (P_2O_5)	20,400	μg l^{-1}
Chloride	82.27	mg l^{-1}
Magnesium	102.35	mg l^{-1}
Calcium	253.78	mg l^{-1}
Potassium	202.70	mg l^{-1}
Boron	0.98	mg l^{-1}
Copper	0.978	mg l^{-1}
Zinc	0.58	mg l^{-1}
Iron	2.33	mg l^{-1}
Manganese	0.95	mg l^{-1}

calculated on a yearly basis, normalised again to the Functional Unit. For delivery to the customers, an average distance was calculated based on current distribution from the farm. As with the transport of inputs and transport of waste, an average delivery mass was calculated, multiplied by the average distribution distance to give the kg km^{-1} for the delivery of lettuce. Produce is exclusively delivered by electric bicycle courier within the city in which the vertical farm is located.

2.4.6. Alternative scenarios

As noted above, alternative scenarios to the current VF practice are examined in this research. The UK's strategy to reach net zero has set targets for all electricity to come from 100% zero-carbon sources by 2035, with the UK national grid electricity set to be mostly renewable energy powered (BEIS, 2021). Current UK electricity average grid mix (derived from the Digest of UK Energy Statistics, 2022), and National Grid projections for 2030, 2040 and 2050 are modelled to see the impact on VF emissions. Another alternative scenario is the recirculation of water and nutrients, ensuring that there is no nutrient and water loss to the drain from cultivation. Alternative packaging scenarios have been created also, to reflect typical packaging for produce going to general retail (biodegradable polybags and cardboard crates for small and large deliveries respectively). Alternative scenarios for waste management model the disposal of non-recyclable waste via anaerobic digestion or industrial composting. Current practice and the alternative scenarios are summarised in Table 4. More detailed information on modelling the various scenarios is given in the Supplementary Material.

2.4.7. Hotspot analysis

Hotpot analysis was undertaken on the current production system and the scenario utilising current UK electricity average grid mix. This identifies the processes which contribute up to 80% of the impact from the system (Vieira, 2016). The PEF approach was utilised to identify hotspots in the production of VF lettuce across the six chosen impact categories. Further details on hotspot analysis are available in the Supplementary Materials.

2.4.8. Land use

VF systems offer the opportunity to spare land for other potential land use purposes. As noted in section 2.5 the EF 3.0 Land Use indicator is used in the LCIA, but is dimensionless. In order to estimate the potential land area savings from a hypothetical move of the UK market from the field cultivation of lettuce in the UK and Spain into VF systems, UK and Spanish government data for yield and land area used were compared with equivalent data for the case study vertical farm. We note that although lettuce varieties in the field are typically different to those grown in VF systems (head vs looseleaf), they are still comparable due to the near linear relationship between the size of lettuce, the length of time to harvestable size, and as the broadly similar nutritional value between varieties (Gargaro et al., 2023; Kozai and Niu, 2015).

Table 4
Current practice and alternative scenarios evaluated in this study.

Current practice	Alternative scenarios
100% renewable energy	Current UK grid mix (2022) UK 2030 grid mix projection UK 2040 grid mix projection UK 2050 grid mix projection
Water and nutrients go to drain	Water and nutrient recycling scenario
Waste goes to incineration	Anaerobic digestion waste management scenario Industrial composting waste management scenario
Reuseable plastic crate packaging	Single-use cardboard crates and biodegradable polybags

3. Results

3.1. Current VF practice

The LCIA results are given in Fig. 2 and Table 5. In terms of Climate Change impacts (0.74 kg CO_2e kg^{-1} lettuce), electricity demand of the system is the single largest contributor, responsible for 40% of all Climate Change related impacts in the system. The second largest contributor was from the jute fibre plugs which contribute 18% of the overall Climate Change impact. Other processes contributing to Climate Change impact are the incineration of waste (7%) and the plastic inputs (14%). All other processes have less than a 7% contribution to the Climate Change indicator.

Jute fibre plugs and electricity are virtually always the largest contributors to impact across all 6 impact categories (and also all other 22 EF3.0 impact categories). For the Jute fibre plugs, the inputs needed for the production and processing of the fibres drive this high impact across the categories. The Eutrophication Freshwater and Water Use categories are also notable here, with Jute fibre plug production contributing 82% and 44% to these indicator scores respectively. While impact categories such as Water Use had an overall negative environmental impact (0.73 m^3 kg^{-1} lettuce), the processing of wastewater and recycling of plastic and paper waste products rendered some beneficial water use results (-0.04 m^3 kg^{-1} lettuce and -0.016 m^3 kg^{-1} lettuce respectively). A beneficial contribution to the Land Use indicator (12.25 Pt) was also observed from the recycling of paper waste. Transport and distribution impacts had only minor contribution to the environmental impacts of the system, for example contributing less than 2% to the Climate Change score. In the Resource Use Fossil category (9.88 MJ), electricity use and plastic inputs contribute to 32% and 31% respectively, with hydrogen peroxide use also having a substantial impact (19%).

Notable results across the other 22 EF 3.0 impact categories (see Supplementary Material) include electricity production in Resource Use Minerals and Metals, due to the mining impacts associated with renewable energy technologies. NPK inputs (surrogate for Solufeed F in Table 1) also contributed markedly to Inorganic Freshwater Ecotoxicity impacts (17%), yet contributed only 0.3% in Eutrophication Freshwater, its impact overshadowed by the contributions from Jute.

Table 6 summarises the 'hotspot' processes for the six selected impact categories from EF 3.0. Results for the remaining 22 impact categories are available in the Supplementary Material (Fig. S1). The input and output processes included in the hotspot analysis tables contribute to 80% of the impact in at least one of the six selected impact categories.

It should be noted that these LCIA results relate to the primary data and operational aspects of this specific case study vertical farm. It is recognised that not all vertical farms will be, for example, powered by 100% renewable energy or will replicate other aspects of the VF system examined. Hence, the LCA outcomes are reported below for alternative electricity supplies and various waste, water, nutrient, and packaging scenarios.

3.2. Scenarios

3.2.1. Comparison of current practice with current and projected UK electricity grid mix

The comparative environmental impact of cultivation using the vertical farm's current 100% renewable energy and, hypothetically, the use of current and future projected UK electricity grid average mixes is given in Fig. 3. The use of renewable energy is approximately 6 times less impactful in Climate Change than if the current UK electricity average grid mix had been used (0.74 kg CO_2e kg^{-1} lettuce vs 4.32 kg CO_2e kg^{-1} lettuce). The farm's current 100% renewable electricity contributes 38% to the overall Climate Change impact per Functional Unit and this proportion would rise to 89% if current UK electricity average grid mix were used. Fig. 3 illustrates clearly the far higher

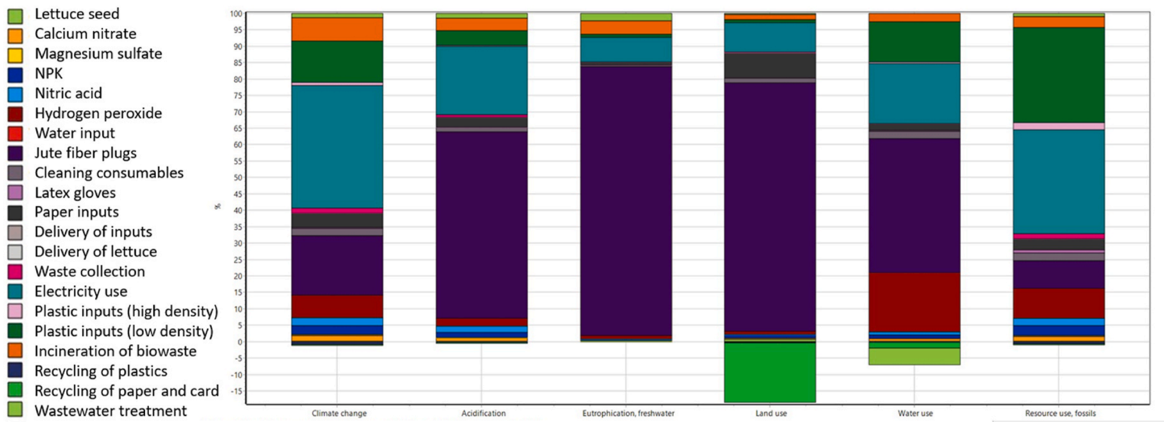

Fig. 2. LCIA results for six selected impact categories (EF 3.0 method), impact per Functional Unit (kg fresh weight of marketable lettuce).

Table 5
LCIA results for six selected impact categories (EF 3.0 method (and ReCiPe 2016 Midpoint (H) V1.1)), impact per Functional Unit (kg fresh weight of marketable lettuce).

Impact category	Climate change	Acidification	Eutrophication, freshwater	Land use	Land use (ReCiPe)	Water use	Resource use, fossils
Unit	kg CO2 eq	mol H+ eq	kg P eq	Pt	m2a crop eq	m3 depriv.	MJ
Total	**7.4E-01**	**8.6E-03**	**1.6E-03**	**1.2E+01**	**2.3E-01**	**7.3E-01**	**9.9E+00**
Lettuce Seeds	8.5E-04	7.9E-06	2.2E-07	1.2E-01	2.9E-03	2.1E-04	1.0E-02
Calcium nitrate	1.3E-02	8.8E-05	2.5E-06	3.9E-02	2.7E-04	6.8E-03	1.4E-01
Magnesium sulphate	1.8E-03	1.1E-05	1.7E-06	6.3E-03	5.2E-05	4.1E-04	3.7E-02
NPK	2.1E-02	1.4E-04	4.6E-06	1.4E-01	8.6E-04	8.4E-03	3.0E-01
Nitric Acid	1.8E-02	1.7E-04	1.7E-06	2.6E-02	2.0E-04	8.0E-03	2.2E-01
Hydrogen peroxide	5.1E-02	2.0E-04	1.8E-05	1.2E-01	9.9E-04	1.4E-01	9.1E-01
Water input	1.2E-05	1.2E-07	4.5E-09	5.7E-05	3.1E-07	1.2E-03	1.6E-04
Jute fibre plugs	1.4E-01	4.9E-03	1.3E-03	1.1E+01	2.2E-01	3.2E-01	8.3E-01
Cleaning consumables	1.7E-02	1.1E-04	5.4E-06	2.2E-01	3.1E-03	1.8E-02	2.5E-01
Latex gloves	2.7E-03	1.8E-05	3.7E-08	4.8E-05	1.5E-07	2.7E-03	8.1E-02
Paper inputs	3.2E-02	2.5E-04	1.6E-05	1.2E+00	1.1E-02	1.7E-02	3.3E-01
Delivery of inputs	1.2E-05	3.5E-08	1.1E-09	9.0E-05	4.9E-07	7.1E-07	1.8E-04
Delivery of lettuce	1.7E-04	1.5E-06	1.1E-07	1.2E-03	6.6E-06	5.0E-05	2.1E-03
Waste collection	1.2E-02	7.2E-05	1.8E-07	2.8E-02	1.9E-04	8.1E-05	1.6E-01
Electricity use	2.8E-01	1.8E-03	1.2E-04	1.3E+00	7.9E-03	1.4E-01	3.2E+00
Plastic inputs (high density)	7.2E-03	2.9E-05	1.2E-06	1.1E-02	6.9E-05	3.3E-03	2.2E-01
Plastic inputs (low density)	9.3E-02	3.8E-04	1.5E-05	1.5E-01	9.4E-04	9.6E-02	2.9E+00
Incineration of biowaste	5.4E-02	3.4E-04	6.6E-05	2.4E-01	8.4E-04	2.0E-02	3.4E-01
Recycling of plastics	−6.7E-03	−1.9E-05	−8.8E-07	−5.2E-02	−2.5E-04	−1.3E-03	−6.6E-02
Recycling of paper and card	−2.5E-03	−2.6E-05	−8.5E-07	−2.7E+00	−2.4E-02	−1.4E-02	−3.0E-02
Wastewater treatment	9.8E-03	1.2E-04	3.5E-05	4.2E-02	1.9E-04	−4.0E-02	9.1E-02

Table 6
Hotspot analysis for the six selected impact categories in the current scenario.

Impact Category	Jute Fibre Plug	Plastic	Electricity	Waste Incineration	Hydrogen Peroxide
Climate Change	18%	14%	**38%**	7%	7%
Acidification	**57%**	5%	21%	–	–
Eutrophication Freshwater	**82%**	–	–	–	–
Land use	**93%**	–	–	–	–
Fossil resource use	–	31%	**32%**	–	19%
Water use	**44%**	14%	20%	–	9%

Bold text indicates the largest single contributor to emissions in each category. A dash indicates the process is not a hotspot.

Climate Change impact of 2022 UK electricity grid average mix than any of the 2030–2050 projected supplies. All the National Grid electricity projections to 2050 have a higher Climate Change impact than the current 100% renewable electricity presently supplied to the vertical farm.

When evaluating the impacts of electricity use in the system from the

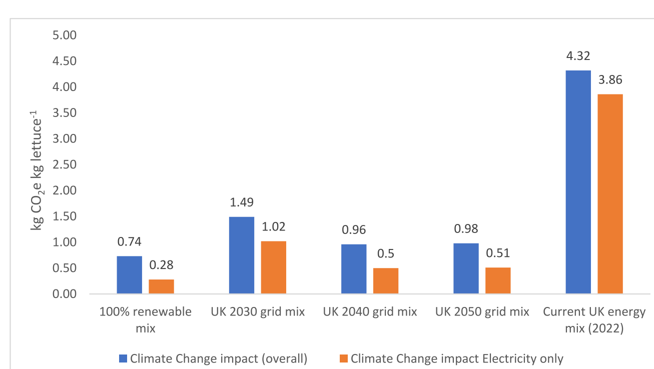

Fig. 3. Climate Change and overall impact of electricity source scenario, compared to current 100% renewable electricity used by the vertical farm per Functional Unit.

2022 UK grid electricity scenario (3.85 kg CO_2e kg^{-1} lettuce), electricity is the dominant contributor to the Climate Change impact of system (89%).

Hotspot analysis on the 2022 UK grid electricity scenario (Table 7) highlights electricity demand as the largest contributor to Climate Change and Fossil Resource Use, with Jute fibre plugs highest in the other four categories. Lighting and HVAC use accounted for 49% (1.87 kg CO_2e kg^{-1} lettuce) and 28% (1.09 kg CO_2e kg^{-1} lettuce) respectively of the overall Climate Change impact.

3.2.2. Comparison of current practice and alternative waste management scenarios

In terms of Climate Change impact from VF current practice, incineration of waste (0.054 kg CO_2e kg^{-1} lettuce) contributes 7% of the overall impact. Recycling of plastic and paper inputs together have a positive environmental impact, due to saving energy and the avoidance of raw material extraction, net saving 0.0092 kg CO_2e kg^{-1} lettuce. For context, waste plastic recycling has a very small positive impact on the environment of −0.0067 kg CO_2e kg^{-1}, and paper recycling −0.0025 kg CO_2e kg^{-1}.

Across the six impact categories, waste plastic recycling has a negative environmental impact. Incineration of waste has a considerable environmental impact, one of the emissions hotspots identified across the 6 highlighted impact categories. In addition to the Climate Change impact, there is a substantial impact on Eutrophication Freshwater (0.0000088 kg P eq kg^{-1} lettuce) and also Acidification (0.000019 kg SO_2 kg^{-1} lettuce).

Waste from the farm is predominantly composed of waste lettuce, Jute plugs and roots, and other non-recyclable waste (plastic film etc.). In this analysis, the effect on environmental impact of potential diversion of waste from incineration to an anaerobic digestor (AD) or to industrial composting was evaluated (Fig. 4). Incineration was the least impactful in terms of Climate Change, yet in terms of other impact categories such as Eutrophication Freshwater and Water Use, it has approx. 55 times greater impact than the next highest waste technology. Finding a circular approach to VF will help reduce the environmental impact of the system, this is explored further in the Discussion. The overall Climate Change impact using AD were approx. 15% higher than incinerating waste (0.84 kg CO_2e kg^{-1} lettuce). Industrial composting on the other hand, was only approx. 3% more impactful (0.76 kg CO_2e kg^{-1} lettuce). These impacts are largely due to the quantity of methane produced in both AD and industrial composting plants (0.0024 kg methane and 0.001 kg methane respectively) compared to incineration (0.00000065 kg methane) per kg of waste processed.

3.2.3. Comparison of current practice and alternative water and nutrient management

At present, the impact from the vertical farm's water use and nutrient disposal are below the threshold above which a trade effluent agreement would be required. Nutrient and chemical impacts are evident across all six selected IF 3.0 impact categories, especially for hydrogen peroxide use (Table 5). In other categories such as Resource Use Minerals and Metals, nutrient and chemical inputs contribute 8% of the impact, insignificant when compared to mineral resources needed for the 80% of impact from the current 100% renewable electricity current supply (modelled on wind production). Considering this, the hydrogen peroxide inputs in the system are the sixth most impactful for Climate Change, higher even than NPK inputs (seventh, 3% of Climate Change impact). Aside from hydrogen peroxide and wastewater processing, there are no other water or nutrient inputs which cause a substantial environmental impact across the six categories (Table 6). Processing of wastewater has an impact of 0.000035 kg P eq kg^{-1} lettuce in Eutrophication Freshwater, but even this only contributes to 2% of the overall impacts in this indicator. Cleaning products also emerged as impactful in the Land Use and Water Use categories, but compared to Jute plugs, their impact inconsequential in comparison. When the system is modelled with current UK grid electricity mix, the environmental impact of nutrient inputs and wastewater treatment have a proportionally lesser impact, but when modelled with exact amounts of nutrients and water necessitated by lettuce crops, the global warming impact of NPK is 0.0000017 kg CO_2e kg^{-1} lettuce, 37% of the current VF system (0.0000046 kg CO_2e kg^{-1} lettuce). Hydrogen peroxide follows the same pattern also, having an impact of 0.0000063 kg CO2e kg^{-1} lettuce, 36% of the current VF system (0.000018 kg CO2e kg^{-1} lettuce).

Water use and wastewater, like nutrient impact, had only minor contribution to the overall impact of the current VF system. Of the six chosen impact categories in Table 5, wastewater processing had a 0.000035 kg P eq kg^{-1} lettuce impact on Eutrophication Freshwater, much lower than Jute plugs at 0.0013 kg P eq kg^{-1} lettuce. For Water Use, the Jute plugs and Hydrogen peroxide had the highest impacts. When modelled with exact quantities of nutrients and water, the water demand is reduced to 37% of the water currently consumed in the VF system. The environmental impact of recirculating water and nutrients and ensuring they do not run to drain is important for increasing the sustainability and circularity of VF and is considered further in the Discussion. Important to note, the current scenario has an overall Climate Change impact of 0.74 kg CO_2 eq kg^{-1} lettuce, marginally more impactful than the metered water and nutrient scenario which had an impact of 0.68 kg CO_2 eq kg^{-1} lettuce.

Table 7
Hotspot analysis for the six selected impact categories in the current UK electricity scenario.

Impact Category	Jute Fibre Plug	Electricity	Hydrogen Peroxide
Climate Change	–	**89%**	–
Acidification	**45%**	37%	–
Eutrophication Freshwater	**84%**	–	–
Land use	**94%**	–	–
Fossil resource use	–	**93%**	–
Water use	**44%**	18%	19%

Bold text indicates the largest single contributor to emissions in each category. A dash indicates the process is not a hotspot.

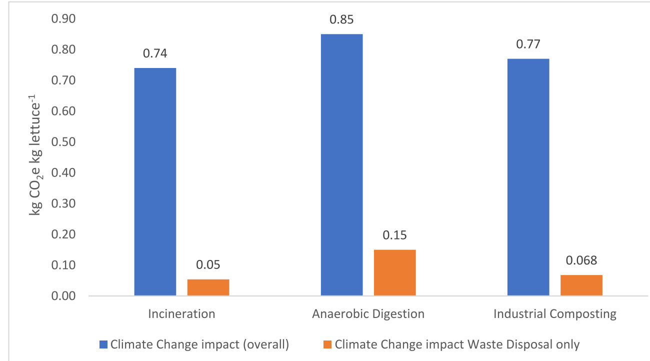

Fig. 4. Climate Change and overall impact of waste management scenarios.

3.2.4. Comparison of current practice and alternative packaging

While packaging use is minimal in the current practice of the vertical farm, the effect of a potential increase in single-use packaging to reflect packaging for produce going to more general retail markets (than the current restaurant customers) was explored. The VF company provided estimates for the equivalent number of cardboard crates for larger deliveries and biodegradable polybags for smaller deliveries that would be needed to fulfil their orders. When the system was modelled using single-use packaging, the Climate Change impact increased to 0.92 kg CO_2 eq kg^{-1} lettuce, 24% higher than the current practice.

Across the other impact categories, the overall impact is similar, except for Land Use (see Supplementary Material for detailed information on this land use), where a 30% increase from 0.23 m^2 kg^{-1} lettuce to 0.30 m^2 kg^{-1} lettuce occurs for land area. This is almost all directly accountable to cardboard crate production, resulting in an increase of

0.07 m^2 kg^{-1} lettuce.

3.2.5. Land area saving

The potential savings in land area when comparing VF and field cultivation for lettuce production in this study are considerable. The total land area required for the production unit of this vertical farm was 150 m^2, with overall land area for all processes on the vertical farm's site occupying 238 m^2. The yield per growing area per year was 97.33 kg m^{-2} yr^{-1}, much higher than the FAO average value for field grown lettuce yield of 1.88 kg m^{-2} yr^{-1} (FAOSTAT, 2022). Based on these yields and current field areas in the UK and Spain for lettuce cultivation for annual UK consumption (8577 ha), we see a potential crop land area saving of 8267 ha, by a hypothetical shift from field cultivation to exclusively vertical farming of lettuces for UK consumption. The land area needed for lettuce production within VF systems in the UK is 310 ha, 3.6% of the current field area needed.

For the current practice of the vertical farm, the Jute fibre plugs account for 93% of the Land Use demand. This suggests that alternative plug material might reduce the Land Use demand still further for the VF and so increase the hypothetical land area saving even more in comparison with field cultivation.

3.2.6. Key findings

- Current practice of the commercial vertical farm in the UK used for the study had a Climate Change impact of 0.74 kg CO_2 eq kg^{-1} lettuce.
- If the vertical farm were to use the 2022 UK electricity average grid mix, rather than its 100% purchase of renewable electricity, the Climate Change impact of the lettuce would increase approx. 6-fold to 4.32 kg CO_2 eq kg^{-1} lettuce.
- Electricity use within VF systems dominated the Climate Change impacts. Lighting and HVAC contribute the majority of this demand (49% and 28% respectively).
- Root plug production (Jute fibre) is the major driver of the vertical farm's impact in Eutrophication Freshwater, Acidification, Land Use and Water Use.
- Water and nutrient level testing are recommended for VF systems. By using metered nutrient concentrations in water, the use of nutrients and water could be reduced by over 60% compared with current practice, and deliver approx. 8% reduction in environmental impact in Climate Change.
- VF of this crop has clear potential to spare very large amounts of land, which could be used for other purposes. If VF were adopted for all UK lettuce production, it would use ~28x less land than current BAU.

4. Discussion

4.1. Climate change and electricity demand

The Climate Change impact result of 0.74 kg CO2 eq kg^{-1} lettuce from the commercial VF system evaluated here is similar to previous values reported on VF systems with renewable energy sources, such as Casey et al. (2022) (0.89 kg CO_2 eq kg^{-1} lettuce), Sandison et al. (2023) (0.48 kg CO_2 eq kg^{-1} lettuce) Martin et al. (2023) (0.75 kg CO_2 eq kg^{-1} lettuce) and Blom et al. (2022) (1.17 kg CO_2 eq kg^{-1} lettuce). In the current case, the use of renewable electricity contributed some 38% to the Climate Change of the lettuce product but this would have risen considerably to 89% if the 2022 UK electricity average grid mix had been used, with the Climate Change impact of the lettuce rising to 4.32 kg CO_2 eq kg^{-1} lettuce. The electricity demand in VF (primarily for lighting and HVAC in the current system) is, unsurprisingly, a major driver of the Climate Change footprint of the system. This is further illustrated by Casey et al.*'s.* work where vertically farmed lettuce using electricity supplied from a relatively clean grid (Norway) and that from a high carbon grid (South Africa) had very different Climate Change impacts at 0.89 kg CO_2 eq kg^{-1} lettuce and 17.8 kg CO_2 eq kg^{-1} lettuce respectively (Casey et al., 2022).

Various UK electricity grid scenarios were evaluated in this research to provide some context to the potential expansion of VF across the UK. Currently, with renewable energy sources, VF presents a comparable environmental footprint to field cultivation and delivery of lettuce. Considering the UK national electricity grid from the present to 2030, 2040 and 2050, the UK grid mix is projected to shift away from the ~45% fossil fuel generation presently on the grid, to as low as 3% of the mix in 2050. With this, the Climate Change impact in 2030 would be reduced to 1.49 kg CO2 eq kg^{-1} lettuce, then rapidly reducing to 0.96 kg CO2 eq kg^{-1} lettuce in 2050, much more comparable to field cultivation. The impacts for the 2050 grid projection, however, remain still higher than if the electricity used was 100% renewable, with impacts something between as low as 0.33 and approx. 0.8 kg CO2 eq kg^{-1} lettuce (findings from Sandison et al., 2023; present study, Martin et al., 2023). The demand for electricity in VF also hold potential for further development ('learning-by-doing', technical and scale advances etc) to reduce consumption from lighting, HVAC and by optimising other processes such as plant density, reducing the heating power required from HVAC by using district heating and cooling systems amongst others (see Pimentel et al., 2023).

A final consideration for VF systems is the extra demand that UK wide deployment of VF systems may have on the grid. The National Grid has already addressed concerns regarding widespread deployment of electric vehicles, stating that the demand for electricity will not be the problem, but that the challenge will be in balancing the load; if too many EVs are charging at once the demand could be too high for the grid. In the UK, if all cars were assumed to be electric, the annual demands based on average miles driven and number of miles per kWh from electric vehicles would be 69.1 TWh, 21% of UK national grid production in 2019 (323 TWh), but the value varies year on year, with capacity in 2005 being 402 TWh, enough to cover EVs and all other electricity (Morris, 2021; National Grid, 2023). VF has the same concerns. The vertical farm in the present study consumes 17.25 kWh per kg of lettuce and the annual electricity demand to meet all lettuce consumed in the UK would be 5.2 TWh. While this is less than 10% of the demand predicted for electric vehicles, and only 2% of the electricity demand from the grid in 2019, there remains a potential risk to grid load if such a technocentric cultivation method were to expand substantially (e.g. for many crops) in the coming years.

4.2. Field cultivation comparison – climate change indicator

Compared with field cultivation impacts, including transportation and distribution, the Climate Change impact of VF systems are similar to those from the field, although with variations. In the general UK context, lettuce is grown in the field from late spring to early autumn, and then in the winter months is mostly imported from southern Spain. The literature presents large differences in Climate Change values for lettuce in field cultivation, depending on how and where the cultivation occurs, and the distances lettuces are transported (Bartzas et al., 2015; Casey et al., 2022; Martin et al., 2023). Casey et al. (2022) found a Climate Change impact of 0.15 kg CO_2 eq kg^{-1} lettuce field grown and delivered in the UK, this being appreciably lower than the 0.68 kg CO_2 eq kg^{-1} lettuce for cultivation in Spain found in their study or for the 0.74 kg CO2 eq kg^{-1} lettuce in VF in the present study. However, Martin et al. (2023) found open field lettuce production in Spain had an impact of 1.6 kg CO_2 eq kg^{-1} lettuce, notably much higher than Casey et al. (2022).

In this study, the impacts from transportation of the vertically farmed lettuce was negligible, contributing to only 0.02% of the overall Climate Change impact. This is unsurprising given the short delivery distances for the crop (average ~11 km), as well as the method of transportation (electric bike courier). Martin et al. (2023) indicate that distribution and delivery to Sweden (from within or abroad, comparable distances to delivery to the UK), add over 0.5 kg CO_2 eq kg^{-1} lettuce to the Climate Change impact from field cultivation. Their study concluded that imported varieties regardless of production method tend to have larger overall Climate Change emissions than VF systems, but again, this is very dependent on the region and electricity source used in both VF and field cultivation (Martin et al., 2023). Sandison et al. (2023) concluded similarly, highlighting that the transportation of field-grown crops from Spain to the UK accounted for 45% of the overall Climate Change impacts (0.56 kg CO_2 eq kg^{-1} lettuce), suggesting lower Climate Change cultivation impact in Spain (~0.31 kg CO_2 eq kg^{-1} lettuce) but higher overall impact than UK cultivation and delivery of UK field grown lettuce (0.46 kg CO_2 eq kg^{-1} lettuce). Future research into field impacts from lettuce cultivation should investigate the impacts from farming on a wider range of soil types and in different climates. For example, a key soil type in the UK for agriculture is on peatland soils, which are commonly used in the UK for cultivation of lettuce and other horticultural crops (~33% of England's vegetables are cultivated in the East Anglian Fenlands). Peat soils are characterised by high carbon content, holding 25% of total global soil carbon (Thers et al., 2023). Estimated annual GHG emissions from arable peat soils are between 26 and 39 t CO_2e ha^{-1} yr^{-1}, which indicate emissions from peatland cultivation of lettuce may be much higher than the estimates from Canals et al. (2009), who investigated lettuce cultivation on mineral soils (Milà Canals et al., 2009; Taft et al., 2017, 2019).

4.3. Non-climate change indicators

Apart from Climate Change impacts - which are heavily influenced by the electricity demand – it is apparent that that the use of Jute fibre for the plugs is a dominant contributor to four of the six impact categories explored in detail. This is a key finding of this research and suggests that the overall environmental sustainability of VF systems could be improved if an alternative medium for supporting roots were utilised. The Jute plugs contributed respectively 57%, 82%, 93% and 44% to the scores for Acidification, Eutrophication Freshwater, Land Use and Water Use (they were also the second highest contributor to Climate Change after electricity demand). Jute fibre is not the only substrate utilised in VF systems and several other substrates such as Hemp, Perlite, Coconut fibre or Rockwool have been used. For example, Martin et al. (2023) evaluated a 50:50 coir-peat mix plug substrate, finding the Climate Change impact to be 0.0026 kg CO_2 kg^{-1} lettuce, very much lower than Jute plugs at 0.013 kg CO_2 kg^{-1} lettuce. While coir is a relatively low impact co-product of coconut production and processing (0.000675 kg CO_2 kg^{-1} coir) peat use has a considerably higher Climate Change impact 0.16 kg CO_2 kg^{-1} peat due to the high carbon emissions associated with its extraction (Pre Consultants, 2021). Future work should explore the environmental impact of different root plugs used for production in VF systems.

The overall impact scores for Acidification, Eutrophication Freshwater and Land-Use are higher for VF when compared with those for field production in Spain. For example, the Eutrophication Freshwater impact in a study based on open field cultivation in Spain was 0.00056 kg P eq kg^{-1} lettuce, an order of magnitude lower than the vertical farm's current practice (0.0016 kg P eq kg^{-1} lettuce). Acidification potential for Spanish field cultivation of lettuce, 0.00066 kg SO_2 eq kg^{-1} lettuce, was again lower than from this study by an order of magnitude (0.0086 kg SO_2 eq kg^{-1} lettuce) (Bartzas et al., 2015). However, Bartzas et al. (2015) did not include the transport of crops in their calculations, which can contribute ~45% of overall impacts (Bartzas et al., 2015; Sandison et al., 2023).

4.4. Towards a more circular economy

Though the electricity demand and Jute fibre are clearly major hotspots of the VF system, further opportunities exist to improve its material circularity and optimise other inputs. For example, water and nutrient consumption may be more extensively recirculated. Implementing environmental monitoring equipment and water nutrient testing equipment could minimise water and nutrients losses to drain by 'flushing' when the electrical conductivity levels drop too low, the current indicator widely used by VF companies. Implementing focussed metering for individual nutrients could enable targeted 'topping up' to maintain the exact amounts of nutrient and other growth solution parameters for the plants, and reduce losses of valuable nutrients that currently remain in wastewater discharged to the drain. However, at present this is not common practice because it remains most cost effective to allow such nutrients to run to drain, rather than to invest in in-house testing equipment. Though the impact from water and nutrient losses is small at present scale, when farms are expanded from the 150 m^2 current facility, the wastewater volume (rather than the nutrient concentrations in the wastewater) may well require further permitting e. g. to a trade effluent agreement. Overall, if the water and nutrient supply were optimised as modelled in the scenario then the Climate Change impact from the system would reduce to 0.68 kg CO_2 eq kg^{-1} lettuce, a reduction of 7% of the impact compared to the vertical farm's current practice.

As noted earlier, the vertical farm's impacts on Eutrophication Freshwater are an order of magnitude higher than field production. However, even when the hypothetical scenario of tightly metered and optimised water and nutrient management was run, the impact in this indicator category still remained higher than field cultivation (0.00056 vs 0.0016 kg P eq kg^{-1} lettuce). The main driver for this is the 84.1% contribution to the system's Eutrophication Freshwater score from use of Jute for plugs (from the field cultivation of the Jute itself). If Jute was replaced by an alternative material with low or zero impact on Eutrophication Freshwater, the impact of the VF system at 0.00025 kg P eq kg^{-1} lettuce would become lower than those for the field cultivation of lettuce in Spain.

This study also evaluated the impact of two alternative waste disposal technologies (AD and composting) for the 'non-recyclable' waste that is currently sent to an incineration with energy from waste (EfW) facility. From the hotspot analysis, incineration of waste (<99% biogenic) contributed only 7% (0.054 kg CO_2 eq kg^{-1} lettuce) of the vertical farm's Climate Change impacts. Comparatively, the impact from AD and composting was 0.15 kg CO_2 eq kg^{-1} lettuce and 0.068 kg CO_2 eq kg^{-1} lettuce, indicating little difference in terms of environmental impact between the three waste management options, with differences likely coming from the variations in the emissions of methane. This LCA has not accounted directly for the potential energy recoverability and displacement of UK grid electricity in its calculations, however the three technologies can produce useful additional products in the form of energy or compost. Incineration of waste has an electricity potential of 0.11 kWh kg^{-1} waste, able to displace grid electricity and render carbon savings of 0.021 kg CO_2e kg^{-1} waste (Dickson et al., 2023). Composting can yield a large quantity of thermal energy, yet energy recovery is limited due to poor heat conduction characteristics of composting feeds (Smith et al., 2017). Industrial composting has non-energy benefits through production of carbon and nitrogen rich fertilisers which can displace fossil fertiliser production. In terms of energy recovery, AD has the highest electricity potential (0.38 kWh kg^{-1} waste), with the biggest potential to displace grid electricity and hence garner carbon savings (0.072 kg CO_2e kg^{-1} waste) (Dickson et al., 2023). AD also produces digestate which, like industrial composting, can be used as a fertiliser (Dickson et al., 2023). The energy production potential and other additional products are worth considering when determining which waste conversion technology is most appropriate environmentally and in the sense of optimising circular economy pathways. AD waste processing does have a higher GHG impact than both incineration and composting, but the energy recoverability and digestate products make it an attractive option. Future work should explore the impacts of different end-of-life technologies with respect to secondary benefits that could be derived from their use.

Finally, we also evaluated the potential impact of single-use packaging in this study. The VF company evaluated currently use reuseable plastic crates for their deliveries, harvesting produce directly into the crates which are emptied by the customers and then returned to the farm. Alternative packaging scenarios have been completed to reflect more typical packaging for produce going to retail. Unsurprisingly, when single-use packaging (cardboard crates and biodegradable plastic bags) is used, a substantial increase occurs in the Climate Change impact from 0.74 to 0.92 kg CO_2e kg^{-1} lettuce. The effective deployment of reuseable crates in the vertical farm's current approach is clearly a better environmental option than moving to single use packaging for its distribution to its current customers (or more widely).

4.5. Land savings

The potential land use savings that VF could provide are considerable and a key area worth exploring further as a potential benefit from VF. Moving cultivation to urban areas where the majority of the UK population resides (84%) (Clark, 2023) can help reduce delivery distances from typical lettuce supply chains, reducing wastage in the distribution of crops, as well as providing fresh produce locally. However, the higher cost of urban real-estate and the potential increase in competition with residential and other commercial buildings has to be considered when making decisions on where to locate vertical farms (Benke and Tomkins, 2017). Transport of crops from southern Spain is typically delivered via refrigerated lorry over 2000 miles distance, and the implementation of more localised crop production would enhance the possibility of greener transport options; via electric bike courier as highlighted in this paper, or via other smaller transport vehicles such as electric vans. As highlighted in Section 3.2.5 Land area saving results, there is a significant land sparing potential from using VF systems across the UK to grow lettuce for the UK population, from 8577 ha to 310 ha. The ability to reduce the land needed primarily comes from the increased yields per area that can be achieved within VF systems. Touliatos et al. (2016) observed yields from VF to be 95 kg lettuce m^{-2}, over 10 times higher than single-layer, conventional hydroponic lettuce production yielding 6.9 kg lettuce m^{-2}. These increased yields per area and hence ability to spare land can provide other benefits that otherwise wouldn't be possible through field cultivation. Spared land could be utilised to cultivate other food crops, an interesting opportunity in the UK because agriculture already utilises 68% of UK land (DEFRA, 2022), and the need to increase food production by at least 70% globally to accommodate a growing population(FAO, 2009). The land could also be used for other purposes to reap environmental benefits and increase ecosystem services by regenerating areas spared from intensive horticulture. This could be via afforestation, rewilding, or bioenergy crop cultivation. Bioenergy feedstock provision especially could present substantial environmental benefits from the remediation of land and increase of soil organic carbon, as well as provision of net-zero energy, and its potential for combination with CCS technology through BECCS, crucial to IPCC projections to limit global warming to 1.5 °C or 2 °C (Almena-Ruiz et al., 2021; Harris and Kountouris, 2020; IPCC, 2018).

This study has assessed lettuce production in a single UK VF and, if expanded to encompass the range of other crops which could be cultivated in VF, further land sparing opportunities could be realised. Life Cycle Assessments, such as those conducted in the present research, are critical to determining the potential environmental pros and cons of VF systems in comparison with field and other cultivation and food provisioning systems of the future.

5. Conclusions

This study has evaluated the environmental impacts of a commercial VF in the UK, and highlights many of the key hotspots from the production (electricity use and jute root plugs) and delivery of an example lettuce crop. Use of renewable energy could help reduce the environmental impact of VF to the point where it is comparable to field production. The work undertaken is one of the first studies globally to utilise primary data from a commercial vertical farm, and the first in the UK context. The outputs set-up a platform for future research; the results have been evaluated in comparison to field lettuce production impacts, highlighting agricultural opportunities that could be had using VF systems, especially with the land sparing potential VF can offer (up to 28x less land needed than current BAU). The results also highlight other areas for future research, for example, exploring the impact from various root plug mediums, optimising electricity use within VF systems, and also how to ensure that the electricity used for VF is decarbonised.

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CRediT authorship contribution statement

Michael Gargaro: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Astley Hastings:** Writing – review & editing, Supervision. **Richard J. Murphy:** Writing – review & editing, Validation, Supervision, Software, Resources, Methodology. **Zoe M. Harris:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2024.143324.

References

Almena-Ruiz, A., Sparks, J., Thornley, P., Roder, M., 2021. BECCS_Briefing_note_final_clean.

Barbosa, G., Gadelha, F., Kublik, N., Proctor, A., Reichelm, L., Weissinger, E., Wohlleb, G., Halden, R., 2015. Comparison of land, water, and energy requirements of lettuce grown using hydroponic vs. Conventional agricultural methods. Int. J. Environ. Res. Publ. Health 12 (6), 6879–6891. https://doi.org/10.3390/ijerph120606879.

Bartzas, G., Zaharaki, D., Komnitsas, K., 2015. Life cycle assessment of open field and greenhouse cultivation of lettuce and barley. Inform. Process. Agric. 2 (3–4), 191–207. https://doi.org/10.1016/j.inpa.2015.10.001.

BEIS, 2021. Plans unveiled to decarbonise UK power system by 2035. BEIS. https://www.gov.uk/government/news/plans-unveiled-to-decarbonise-uk-power-system-by-2035.

Benke, K., Tomkins, B., 2017. Future food-production systems: vertical farming and controlled-environment agriculture. Sustain. Sci. Pract. Pol. 13 (1), 13–26. https://doi.org/10.1080/15487733.2017.1394054.

Birkby, J., 2016. Vertical farming | ATTRA | sustainable agriculture Project. https://attra.ncat.org/htmlpub/vertical-farming/#intro.

Casey, L., Freeman, B., Francis, K., Brychkova, G., McKeown, P., Spillane, C., Bezrukov, A., Zaworotko, M., Styles, D., 2022. Comparative environmental footprints of lettuce supplied by hydroponic controlled-environment agriculture and

field-based supply chains. J. Clean. Prod. 369 https://doi.org/10.1016/j.jclepro.2022.133214.
Clark, D., 2023. Urban and rural population of the United Kingdom from 1960 to 2022. Statista; Statista. https://www.statista.com/statistics/984702/urban-and-rural-population-of-the-uk/.
DEFRA, 2021. United Kingdom food security report 2021: theme 2: UK food supply sources - GOV.UK. https://www.gov.uk/government/statistics/united-kingdom-food-security-report-2021/united-kingdom-food-security-report-2021-theme-2-uk-food-supply-sources.
DEFRA, 2022. Agricultural land use in United Kingdom at 1 june 2022. DEFRA. https://www.gov.uk/government/statistics/agricultural-land-use-in-the-united-kingdom/agricultural-land-use-in-united-kingdom-at-1-june-2022.
Dias, G.M., Ayer, N.W., Khosla, S., Van Acker, R., Young, S.B., Whitney, S., Hendricks, P., 2017. Life cycle perspectives on the sustainability of Ontario greenhouse tomato production: Benchmarking and improvement opportunities. J. Clean. Prod. 140, 831–839. https://doi.org/10.1016/j.jclepro.2016.06.039.
Dickson, E.M., Hastings, A., Smith, J., 2023. Energy production from municipal solid waste in low to middle income countries: a case study of how to build a circular economy in Abuja, Nigeria. Front. Sustain. 4 https://doi.org/10.3389/frsus.2023.1173474.
EC, 2023. Food crises worsen in the wake of Russia's invasion of Ukraine, economic shocks resulting from COVID-19 and weather extremes. European Commission. https://international-partnerships.ec.europa.eu/news-and-events/news/food-crises-worsen-wake-russias-invasion-ukraine-economic-shocks-resulting-covid-19-and-weather-2023-05-03_en.
EC-JRC, 2018. Environmental footprint reference package 3.0 (EF 3.0). European Commission- JRC. https://eplca.jrc.ec.europa.eu/LCDN/developerEF.html.
Ecoinvent, 2022. Ecoinvent database. Ecoinvent. https://ecoinvent.org/the-ecoinvent-database/#!/dqg.
English, A., Hughes, I., 2022. POLICY BRIEF 4: How can Policymakers Boost Fruit and Vegetable Production and Consumption? Food Foundation.
FAO, 2009. Global Agriculture towards 2050.
FAOSTAT, 2022. Faostat. Fao. https://www.fao.org/faostat/en/#data/QCL.
Fazio, S., Biganzioli, F., De Laurentiis, V., Zampori, L., Sala, S., Diaconu, E., 2018. Supporting information to the characterisation factors of recommended EF life cycle impact assessment methods – Version 2, from ILCD to EF 3.0. Publications office. https://doi.org/10.2760/002447.
Finkbeiner, M., Inaba, A., Tan, R., Christiansen, K., Klüppel, H.-J., 2006. The new International standards for life cycle assessment: ISO 14040 and ISO 14044. Int. J. Life Cycle Assess. 11 (2), 80–85. https://doi.org/10.1065/lca2006.02.002.
Gargaro, M., Murphy, R.J., Harris, Z.M., 2023. Let-us investigate; A Meta-analysis of influencing factors on lettuce crop yields within controlled-environment agriculture systems. In: Plants, vol. 12. Multidisciplinary Digital Publishing Institute (MDPI). https://doi.org/10.3390/plants12142623. Issue 14.
Harris, Z.M., Kountouris, Y., 2020. Vertical farming as a Game changer for BECCS technology deployment. Sustainability 12 (19), 8193. https://doi.org/10.3390/su12198193.
IPCC, 2018. Global warming of 1.5°C. https://www.ipcc.ch/site/assets/uploads/sites/2/2019/06/SR15_Full_Report_High_Res.pdf.
IPCC, 2022. Food security. In: Climate Change and Land. Cambridge University Press, pp. 437–550. https://doi.org/10.1017/9781009157988.007.
ISO, 2006a. ISO - ISO 14040:2006-Environmental management — life cycle assessment — principles and framework. ISO. https://www.iso.org/standard/37456.html.
ISO, 2006b. ISO - ISO 14044:2006-Environmental management — life cycle assessment — requirements and guidelines. ISO. https://www.iso.org/standard/38498.html.
Kozai, T., Niu, G., 2015. Plant factory as a resource-Efficient closed plant production system. Plant Factory: Indoor Vertic. Farm. Syst. Eff. Qual. Food Prod. 69–90. https://doi.org/10.1016/B978-0-12-801775-3.00004-4.
Martin, M., Elnour, M., Siñol, A.C., 2023. Environmental life cycle assessment of a large-scale commercial vertical farm. Sustain. Prod. Consum. 40, 182–193. https://doi.org/10.1016/j.spc.2023.06.020.
Milà Canals, L., Muñoz, I., Hospido, A., Plassmann, K., McLaren, S., Edwards-Jones Project Manager, G., Hounsome, B., Almudena, Muñoz, Hospido Katharina Plassmann Sarah McLaren, I., 2009. LIFE CYCLE ASSESSMENT (LCA) OF DOMESTIC VS. IMPORTED VEGETABLES. Case studies on broccoli, salad crops and green beans Life Cycle Assessment (LCA) of Domestic vs. Imported Vegetables. Case studies on broccoli, salad crops and green beans Llorenç Milà i Canals. http://www.surrey.ac.uk/CES.
Morgan, S., Fuglie, K., Jellife, J., 2022. World Agricultural Output Growth Continues to Slow, Reaching Lowest Rate in Six Decades. USDA. https://www.ers.usda.gov/amber-waves/2022/december/world-agricultural-output-growth-continues-to-slow-reaching-lowest-rate-in-six-decades/.
Nicholson, C.F., Harbick, K., Gómez, M.I., Mattson, N.S., 2020. An economic and environmental comparison of conventional and controlled environment agriculture (CEA) supply chains for Leaf lettuce to US cities. In: Food Supply Chains in Cities, pp. 33–68. https://doi.org/10.1007/978-3-030-34065-0_2.
Orsini, F., Pennisi, G., Zulfiqar, F., Gianquinto, G., 2020. Sustainable use of resources in plant factories with artificial lighting (PFALs). Eur. J. Hortic. Sci. 85 (5), 297–309. https://doi.org/10.17660/EJHS.2020/85.5.1.
Pre Consultants, 2021. SimaPro 9.3.0.3 (9.3.0.3). Pre consultants. https://simapro.com/wp-content/uploads/2021/12/SimaPro930WhatIsNew.pdf.
Rangathan, J., Waite, R., Searchinger, T., Hanson, C., 2018. How to sustainably feed 10 billion people by 2050, in 21 Charts. World Resour. Instit. https://www.wri.org/insights/how-sustainably-feed-10-billion-people-2050-21-charts.
Sandison, F., Yeluripati, J., Stewart, D., 2023. Does green vertical farming offer a sustainable alternative to conventional methods of production?: a case study from Scotland. Food Energy Secur. 12 (2) https://doi.org/10.1002/fes3.438.
Smith, M.M., Aber, J.D., Rynk, R., 2017. Heat recovery from composting: a Comprehensive review of system Design, recovery rate, and Utilization. In: Compost Science and Utilization, vol. 25. Taylor and Francis Inc, pp. S11–S22. https://doi.org/10.1080/1065657X.2016.1233082.
Taft, H.E., Cross, P.A., Edwards-Jones, G., Moorhouse, E.R., Jones, D.L., 2017. Greenhouse gas emissions from intensively managed peat soils in an arable production system. Agric. Ecosyst. Environ. 237, 162–172. https://doi.org/10.1016/j.agee.2016.11.015.
Taft, H.E., Cross, P.A., Hastings, A., Yeluripati, J., Jones, D.L., 2019. Estimating greenhouse gases emissions from horticultural peat soils using a DNDC modelling approach. J. Environ. Manag. 233, 681–694. https://doi.org/10.1016/j.jenvman.2018.11.113.
Thers, H., Knudsen, M.T., Lærke, P.E., 2023. Comparison of GHG emissions from annual crops in rotation on drained temperate agricultural peatland with production of reed canary grass in paludiculture using an LCA approach. Heliyon 9 (6), e17320. https://doi.org/10.1016/j.heliyon.2023.e17320.
Touliatos, D., Dodd, I.C., Mcainsh, M., 2016. Vertical farming increases lettuce yield per unit area compared to conventional horizontal hydroponics. Food Energy Secur. 5 (3), 184–191. https://doi.org/10.1002/fes3.83.
United Nations, 2019a. Around 2.5 billion more people will be living in cities by 2050, projects new UN report | United Nations. United Nations Depart. Econ. Soc. Aff. https://www.un.org/en/desa/around-25-billion-more-people-will-be-living-cities-2050-projects-new-un-report.
United Nations, 2019b. Growing at a slower pace, world population is expected to reach 9.7 billion in 2050 and could peak at nearly 11 billion around 2100 | UN DESA | United Nations Department of Economic and Social Affairs. United Nations Depart. Econ. Soc. Aff. https://www.un.org/development/desa/en/news/population/world-population-prospects-2019.html.
USDA, 2023. International Agricultural Productivity.
Vieira, M., 2016. Hotspot analysis in the PEF approach. Pre-Sustainability- PEF. https://pre-sustainability.com/articles/pef-series-hotspot-analysis-in-the-pef-approach/.
Webb, N.P., Marshall, N.A., Stringer, L.C., Reed, M.S., Chappell, A., Herrick, J.E., 2017. Land degradation and climate change: building climate resilience in agriculture. Front. Ecol. Environ. 15 (8), 450–459. https://doi.org/10.1002/fee.1530.